Reconstruction-Free Evaluation for Dead Sea Scroll Restoration: Token Alignment, Transcription-Encoded Evidence, and Unknown-Length Gaps

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Track: Computational Humanities & Ancient Language Restoration

Abstract

Dead Sea Scroll lacunae differ from ordinary masked-word tasks: visible letters may survive, while character length, word count, and boundaries remain uncertain. We introduce a reconstruction-free corpus derivative and two complementary evaluations that keep these information sources explicit. Across three matched training seeds and 93 natural single-word lacunae, a preserved-only masked language model averages 14.7% Top-10 agreement with attributed proposals from context alone and 56.6% with development-tuned soft trace evidence; a frequency baseline with hard traces reaches 36.6%. On 300 non-overlapping synthetic spans sampled across 79 held-out scrolls, the strongest unknown-length decoder reaches 7.7% exact Top-10 and recovers no three-word target. The practical verdict is explicit: the method is useful as a trace-aware candidate ranker for eligible single-word lacunae, but it is not an effective general restoration system for unknown-length multiword gaps. Agreement with published proposals is not historical ground truth.

1 Introduction

The Dead Sea Scrolls preserve ancient Hebrew and Aramaic in manuscripts damaged by abrasion, tears, decay, and uncertain joins. The data are not dominated by isolated missing words: in our reconstruction-free transcription derivative, only 18.8% of lacuna records occupy one source-word position, the median occupies four, and 32.7% occupy six or more. A transcription may encode partial letters and an editor's estimate of missing extent, while the original word count and boundaries remain uncertain. A restoration system must therefore state exactly which evidence it receives.

Masked language models (MLMs) provide contextual predictions, but their tokenization can leak or obscure gap structure. A WordPiece mask normally specifies a token count, while an epigrapher observes an uncertain region measured in material space. Character models offer direct alignment, yet lexical models may rank complete words more effectively. Retrieval can expose parallels but can also leak the target or reproduce editorial traditions. Evaluation against a modern reconstruction is similarly ambiguous: it measures agreement with an edition, not recovery of inaccessible historical truth.

We organize these issues into two complementary tasks: (A) synthetic damage to preserved text with a known answer and (B) agreement with multiple attributed readings at natural lacunae. Task A hides character length, word count, and boundaries and uses exact complete-span Top-10. Task B separates context-only, visible-trace, and editor-length conditions on one fixed denominator.

The present draft makes four evidence-bounded contributions:

- a reconstruction-free non-biblical corpus derivative and a reproducible profile of 27,814 transcription-derived lacuna records;
- a shared-denominator benchmark that separates context, transcription-visible letters, and editor-derived length rather than collapsing them into a "physical" condition;
- a paired 93-target ablation with three matched MLM seeds, soft and hard trace conditions, non-neural baselines, hierarchical seed-and-scroll uncertainty, and composition-exclusion sensitivity; and
- a manuscript-balanced 300-span benchmark showing that unknown-length multiword decoding remains unresolved.

The contribution is therefore primarily methodological and diagnostic rather than a claim of historically correct restoration.

2 Related Work

Ancient-text restoration. Pythia introduced neural character restoration for Greek inscriptions and reported character error rate (CER) and Top-20 sequence recovery [1]. Recurrent models were also applied to fragmentary Babylonian texts [7]. Lazar et al. formulated Akkadian missing-sign completion as MLM inference and included expert evaluation [10]. Ithaca combined character and word representations with attribution, dating, saliency, and a controlled historian study [2]. Aeneas subsequently integrated images, transcriptions, parallel retrieval, arbitrary-length restoration, and historian evaluation for Latin inscriptions [3]. The field-wide survey of Sommerschield et al. emphasizes that restoration must be located within a broader scholarly workflow [17].

Semitic and manuscript models. Embible ensembles character and word transformers for synthetically masked Biblical Hebrew and Aramaic [9]. MsBERT targets orthographic variation and lacunae in post-biblical Hebrew

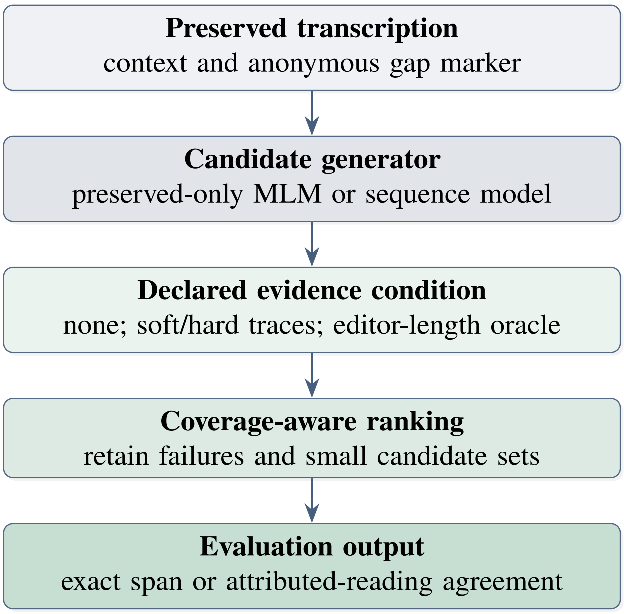

Figure 1: Evaluation architecture. Visible letters are copied from a transcription; no image or stroke inference is performed. Each result names its supplied evidence.

manuscripts [16]. TavBERT studies character-level pretraining for morphologically rich languages [13]; AlephBERT and DictaBERT provide strong Hebrew subword baselines [14, 15]. These models differ in period, register, tokenizer, and pretraining data, so a DSS comparison requires a shared target set and information condition.

Real lacunae and material evidence. The MAAT corpus preserves lacunae, restorations, and editorial structure from EpiDoc instead of reducing every example to random masking [8]. Levine et al. evaluate ranked, variable-length completions for Coptic manuscripts and explicitly frame predictions as scholar assistance [11]. For the DSS, Uzan et al. train a modified PixelCNN on infrared crops from 11Q5, evaluate synthetic fragment-edge masks, and inspect 19 real broken letters [19]. Their visual task is complementary to our transcription-level ranking, but its single-scroll, individual-letter setting does not directly yield word candidates. Our "visible traces" are transcription-mediated; claims about ink or strokes require registered image inputs.

3 Tasks and Information Conditions

Let $x_{<g}$ and $x_{>g}$ be preserved context around a gap g. A system returns an ordered set $C_k(g)$ of at most k complete strings. For synthetic damage with known preserved target y_g,

$$H_k(g) = \mathbb{I}[y_g \in C_k(g)]. \quad (1)$$

For a natural lacuna with a set Y_g of distinct, bibliographically attributed proposals,

$$A_k(g) = \mathbb{I}[C_k(g) \cap Y_g \neq \varnothing]. \quad (2)$$

A_k is *literature agreement*; neither inclusion nor exclusion establishes truth.

Table 1: Reconstruction-free non-biblical corpus profile. Held-out refers to the checkpoint-associated split used by the preserved-only models.

Measure	Full corpus	Held-out
Scroll identifiers	736	220
Training/evaluation chunks	1,647	407
Preserved word tokens	95,736	25,313
Lacuna records	27,814	6,814
Damaged word positions	165,239	43,401

We distinguish five information conditions:

- CONTEXT: linguistic context only.
- SOFTVISIBLE: a development-fitted penalty for disagreement with visible segments and anchored edges; candidates are reranked but never discarded.
- VISIBLE: candidates must contain the Hebrew segments shown outside reconstruction brackets, respecting anchored edges when encoded.
- EDITORLENGTH: candidate length must fall within one character of the QD display/initial length. This is an oracle/editor-assisted diagnostic.
- VISIBLE+LENGTH: the conjunction of the preceding filters, also oracle-assisted because of length.

Visible traces are transcription evidence, not independent observations collected by our system. The current pipeline neither reads multispectral images nor estimates stroke probabilities.

4 Data

Corpus construction. The corpus is derived from the ETCBC DSS Text-Fabric dataset [6]. Biblical scrolls are excluded. Editorial reconstruction text is never emitted as a training label: physically preserved characters remain visible and gaps become anonymous <GAP> tokens. Deterministic scroll-level partitions contain 1,002 training, 238 development, and 407 held-out chunks. Corpus and lacuna files are recorded with SHA-256 hashes. Table 1 reports counts regenerated directly from the validated derivative.

Scroll eligibility is task-specific. An archival registry and a modeling population answer different questions. We retain all 736 non-biblical scroll identifiers for provenance, including fragments that contribute only damage metadata. A scroll enters primary MLM text only if it yields at least one 128-unit chunk with 20 preserved words. Consequently, 377 scrolls contribute model text and 359 do not (Table 2). The latter are not deleted: 25 contain no strictly preserved words, while 71 satisfy the predeclared extreme-fragmentation flag of fewer than 20 preserved source words and a preserved-position fraction below 10%. These overlapping groups remain auditable but are excluded from primary linguistic modeling and contextual-restoration target selection. Neither evaluation contains an extreme-fragmentation scroll. Preservation fraction here is a property of classified transcription positions, not a measurement of surviving parchment area.

Table 2: Task-specific scroll eligibility. Categories overlap: zero-preserved and extreme-fragmentation scrolls are subsets of the no-chunk population. Preservation fraction is a transcription-position proxy, not parchment-area intactness.

Population	Scrolls	Treatment
Archival registry	736	retain for provenance
Primary MLM contributors	377	at least one eligible chunk
No primary MLM chunk	359	exclude from MLM text
Zero preserved words	25	exclude from linguistic modeling
Extreme fragmentation	71	flag; exclude from primary context task

Table 3: Lacunae by estimated source-word positions. Counts derive from consecutive damaged positions after editorial text is removed; they are structural estimates, not direct measurements of a single physical hole.

Positions	Full corpus	Held-out
1	5,239 (18.8%)	1,207 (17.7%)
2	4,536 (16.3%)	1,090 (16.0%)
3	3,997 (14.4%)	962 (14.1%)
4–5	4,941 (17.8%)	1,186 (17.4%)
6+	9,101 (32.7%)	2,369 (34.8%)

This policy avoids a global intactness blacklist. A globally damaged manuscript may contain a locally usable island; such material may enter a separately reported preserved-run ablation if it satisfies reconstruction-free local criteria, but it cannot silently enter the primary population. Conversely, a high scroll-wide preservation rate cannot rescue a target with inadequate local left/right context. Target eligibility is therefore determined at the lacuna as well as the scroll level.

What does a lacuna look like? The 27,814 records contain 165,239 damaged source-word positions (Table 3). Their median span is four positions, the 90th percentile is 13, and the maximum is 268. These are consecutive source-word positions after editorially supplied text has been removed. They describe the structure of the transcription and must not be read as 27,814 independent, metrically measured physical holes.

The transcription nevertheless preserves useful evidence. At least one word retains a visible trace in 23,632 lacuna records (85.0%), but only 49,130 of the 165,239 damaged word positions (29.7%) retain any trace. An estimated missing-character extent is available for 25,509 records (91.7%); its median is five characters and its nearest-rank interquartile range is 2–12. Both left and right preserved context are available for 26,467 records (95.2%), with a median of 14 total context words. The contrast between record-level evidence and position-level sparsity motivates constrained ranking without assuming that every hidden word has a trace.

Two evaluation views. No single benchmark answers both whether a system can recover a known string and whether it can surface a defensible proposal at a genuine lacuna. We therefore use the complementary views in Table 4: synthetic damage supplies an observable answer under controlled masking, while the QD set preserves natural ambiguity and supports only literature-agreement claims.

Table 4: The two principal evaluation views. Synthetic damage has an observable preserved answer; QD measures agreement with attributed proposals and not historical truth.

Set	Targets	Scrolls	Reference
Unknown-length synthetic	300	79	preserved complete span
QD natural lacunae	93	40	184 attributed readings

Synthetic alignment diagnostic. The shared 729-word diagnostic masks intact content words and tests whether candidate generation aligns with the same character interval. It is useful for tokenizer coverage, but it does not simulate unknown-length natural damage and is not the primary benchmark.

Unknown-length spans. We hide 300 preserved DSS spans, balanced across one, two, and three words, after fitting decoding parameters on 60 development spans. Character length, word count, and boundaries are hidden. To prevent large compositions from dominating, deterministic round-robin sampling cycles through manuscripts and rejects overlapping hidden intervals. The test covers 79 held-out scrolls (77, 55, and 41 in the respective word-count strata) and uses two context words on each side.

Qumran Digital targets. The QD snapshot dated 21 May 2026 contains 1,811 publication rows at 346 candidate locations. Predeclared structural and context filters retain 93 non-biblical single-word targets across 40 scrolls and 184 distinct attributed readings. Crucially, a proposal is not removed merely because it conflicts with a tested evidence filter; it remains a possible reference and becomes unretrievable in that condition. All 93 targets are held out under the frozen manifest used for checkpoint training. QD selected disputed sites and is working scholarly data; the benchmark is neither random nor exhaustive.

5 Models and Decoding

Word-span baseline. The word-span baseline enumerates one to three word-mask hypotheses and ranks complete surfaced strings. It is fine-tuned only on preserved non-biblical text. The tokenizer can represent 81.7% of complete synthetic targets with one token per word; all other targets remain in the denominator as misses.

Character and byte models. TavBERT supplies character-aligned masks and ByT5 supplies a byte-level sequence-to-sequence comparison [12]. Character alignment avoids WordPiece drops, but alignment alone does not imply superior exact-span ranking. Three ByT5 checkpoints vary only the training seed (41–43) and share three epochs, batch size 32, learning rate 10^{-3}, and the same corpus hash.

Embible-style combinations. We implement both the paper-described overlap rule and a development-fitted rank fusion of word and character candidates. Under the locked tie-break, a simpler system is retained when exact Top-10 ties. This prevents selecting a more complex ensemble on secondary metrics alone.

Frequency baseline. For the QD task, we rank Hebrew surface forms by frequency in preserved training chunks, then apply exactly the same trace and length filters as the MLM. This baseline isolates how much apparent success follows from a narrow constraint set rather than contextual modeling.

Soft traces and exact-context retrieval. Soft trace cost is the visible-string longest-common-subsequence deficit plus anchored-edge disagreement. Its weight is selected from a fixed grid on 300 preserved development words using only QD trace shapes, never QD readings. A non-neural contextual baseline ranks exact preserved-training context matches and backs off to frequency; it uses at most two words per side and never crosses a natural <GAP>.

6 Evaluation Protocol

The primary automatic task is exact complete-span Top-10 under unknown length, word count, and word boundaries. Exact Top-1/5/20, MRR, CER, boundary F1, word-count mean absolute error, decoder failure, calibration, selective accuracy, and latency are secondary. Slot-level word hits are diagnostic only.

The frozen split is scroll-disjoint. Test data are never used for selection. Failed decoding, tokenizer misalignment, unrepresentable targets, empty candidate sets, and lists shorter than K remain in the denominator. Exact duplicate normalized chunks are absent across the development/test boundary. Maximum preserved-word five-gram Jaccard is 0.174; none of 407 held-out chunks reaches 0.5. Because 26 of 88 composition labels cross partitions, we separately report the QD targets whose composition labels are unseen in train and development.

We report 95% bootstrap intervals clustered by scroll and paired cluster intervals for evidence-condition deltas. For the QD MLM, three checkpoints vary only seed (41–43), use two epochs, batch size 16, learning rate 3×10^{-5}, and the same 1,002 training chunks; hierarchical intervals resample both seeds and scrolls. Target-level predictions, exclusions, manifests, and code/data/model/result hashes accompany the aggregate tables. These choices follow reproducibility guidance in NLP [5, 18].

Protocol validity and scope. Reconstruction-free labels, scroll-disjoint splits, fixed denominators, non-neural controls, matched seeds, and clustered uncertainty make this protocol suitable for testing candidate ranking under declared transcription evidence. They do not validate historical truth, physical ink interpretation, or unconstrained restoration of arbitrary lacunae. We therefore judge the method separately on the natural single-word assistance task and the harder unknown-length synthetic task.

Table 5: Shared synthetic alignment diagnostic (n = 729 preserved content words). Hit rates use all targets as the denominator; alignment failures are misses.

System	Hit@1	Hit@10	Drops
TavBERT base (character)	7.27%	21.12%	0.00%
TavBERT FT (character)	7.96%	20.58%	0.00%
MsBERT FT (WordPiece)	6.04%	13.31%	38.27%
ByT5 unified (byte)	0.82%	5.35%	0.00%

Table 6: Unknown-length synthetic damage on 300 non-overlapping preserved DSS spans sampled across 79 held-out scrolls. Intervals use 10,000 manuscript-cluster resamples. The oracle uses unavailable gold word length.

System	Top-1	Top-5	Top-10	95% CI	CER
Preserved-only word span	4.0%	6.7%	7.7%	5.0–10.8%	0.893
Preserved-only TavBERT	1.0%	2.3%	2.7%	0.9–5.1%	0.898
Embible-style overlap	2.3%	2.3%	2.3%	0.6–4.6%	0.884
Dev-fitted rank fusion	4.3%	7.0%	7.7%	5.1–10.6%	0.868
Oracle word-length filter	6.7%	10.7%	12.0%	–	0.860

7 Results

The results follow the task's sources of uncertainty rather than a leaderboard: we first test whether tokenization can align a known interval, then require exact recovery when length and boundaries are hidden, and finally measure agreement with attributed proposals at natural lacunae. Metrics are comparable only within each table.

Alignment is necessary but insufficient. Table 5 uses a genuinely shared denominator. TavBERT base obtains the highest Hit@10, while DSS fine-tuning does not improve it. MsBERT's aligned-only rates are 9.78% Hit@1 and 21.56% Hit@10, but presenting those alone would condition on successful tokenization. With 279 of 729 failures counted as misses, its all-target rates are 6.04% and 13.31%. DictaBERT-char is omitted because the available 22.80% result comes from a different 250-word pilot.

Unknown-length multiword restoration remains open. The word-span model and rank fusion tie at 7.7% exact Top-10 (Table 6); the simplicity rule retains the word model. The oracle word-length filter reaches 12.0%, quantifying the benefit of unavailable structural information rather than a deployable gain. Table 7 exposes the decisive weakness: the baseline falls from 22.0% on one-word spans to 1.0% on two-word spans and 0.0% on three-word spans. Thus the benchmark rejects, rather than validates, a claim of solved unknown-length restoration.

Three matched ByT5 checkpoints were rerun on the identical target manifest. Their overall exact Top-10 scores are 1.3%, 1.3%, and 1.0%; every checkpoint obtains 0% on the two- and three-word strata. The seed result therefore

Table 7: Exact Top-10 and representability by hidden word count. Each stratum has 100 spans. Coverage is diagnostic; uncovered cases remain misses.

Measure	1 word	2 words	3 words
Word-span exact Top-10	22.0%	1.0%	0.0%
Complete-span coverage	91.0%	77.0%	77.0%

Table 8: Agreement with any attributed reading at 93 Qumran Digital targets. The soft-trace MLM Top-10 seed-and-scroll hierarchical bootstrap 95% CI is 46.5–66.7. All rows retain the same targets and references.

Condition	Top-1	Top-5	Top-10	Top-20
MLM: context only (3-seed mean)	5.0%	11.1%	14.7%	16.8%
Frequency: context only	0.0%	2.2%	3.2%	6.5%
MLM: soft traces (3-seed mean)	35.1%	53.4%	56.6%	61.3%
MLM: hard traces (3-seed mean)	32.6%	49.5%	52.3%	55.2%
Frequency: visible traces	22.6%	32.3%	36.6%	36.6%

confirms a stable negative finding rather than a competitive sequence baseline.

Soft transcription traces carry most of the QD signal. On the same 93 targets, mean MLM Top-10 agreement changes from 14.7% context-only to 56.6% with soft traces (Table 8). The paired improvement is 41.9 points, with a 30.8–53.0 seed-and-scroll hierarchical 95% interval; the absolute soft-trace interval is 46.5–66.7%. The hard-trace mean is 52.3% and varies more across seeds (SD 2.2 versus 0.6 points). Frequency with hard traces reaches 36.6%; the soft-MLM advantage is 20.1 points (9.2–30.2). Exact-context retrieval also reaches 36.6% and therefore adds no non-neural gain. The seed-42 editor-length proxy adds no Top-10 gain over hard traces and remains an oracle diagnostic.

Trace-count error analysis. Visible conditioning is not uniformly beneficial. Averaged across seeds, the 12 one-letter targets score 41.7% context-only, 33.3% with soft traces, and 16.7% with hard filtering. For 14 two-letter targets, soft and hard traces reach 64.3% and 66.7%; for 67 targets with at least three letters they reach 59.2% and 55.7%. Soft evidence therefore repairs much, but not all, of the brittle one-letter failure without discarding candidates.

Composition, context, calibration, and memorization checks. The frozen split is manuscript-disjoint but not composition-disjoint. We therefore train a matched seed-42 sensitivity checkpoint after removing every training scroll whose non-empty composition label occurs among QD targets, retaining 785 of 1,002 chunks and zero such label overlap. On the 68 labeled QD targets, regular and excluded training have identical Top-10 hit rates: 11.8% context-only, 52.9% soft-trace, and 50.0% hard-trace, although rankings differ. This is a single-seed, reduced-data sensitivity rather than a representative new split.

With seed 42, context-only Top-10 rises from 6.5% at two words per side to 11.8%, 14.0%, and 15.1% at five, ten, and

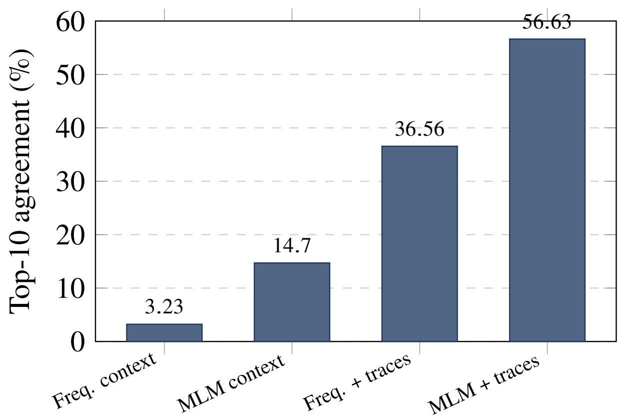

Figure 2: QD Top-10 literature agreement on a fixed 93-target denominator. MLM bars are three-seed means; the final bar uses soft trace evidence.

full stored context; soft-trace Top-10 remains 55.9–57.0%. Among 80 nonempty hard-trace candidate sets, retaining the 50% with highest normalized Top-1 score gives 60.0% Top-1 literature agreement versus 38.8% at full nonempty coverage; this is agreement calibration against incomplete proposals, not truth calibration. Finally, 59 targets have an attributed word seen somewhere in preserved training text, but none has an exact preserved-training trigram or five-gram containing that reading. Base-model pretraining exposure cannot be excluded.

Operational verdict. For eligible natural single-word lacunae, the soft-trace MLM is useful: its Top-10 literature agreement exceeds the hard-trace frequency control by 20.1 points, with a seed-and-scroll interval that remains positive. Context-only ranking is weak, so the gain cannot be attributed to the language model alone. For unknown-length spans, the verdict is negative: 7.7% exact Top-10 overall, 1.0% for two words, and 0% for three words are not adequate for general restoration. The present method should therefore prioritize candidates under explicit transcription evidence, not autonomously fill arbitrary gaps.

8 Comparative Synthesis

Table 9 places the framework in its lineage. Lazar establishes ancient Semitic MLM completion; Embible motivates granularity ensembles; MsBERT motivates manuscript-domain pretraining; Aeneas demonstrates arbitrary-length multimodal restoration. Our contribution is a DSS corpus and evaluation framework that separates supplied evidence from model ability; we do not claim a new state-of-the-art architecture.

9 Reproducibility and Data Statement

Core paper-facing numbers are regenerated from frozen result snapshots containing target-manifest, model, input,

Table 9: Methodological synthesis. Scores are not compared across corpora. A dash indicates that a dimension is not central to the cited evaluation.

Dimension	Lazar (2021)	Embible (2024)	MsBERT (2024)	Aeneas (2025)	This work
Material/language	Akkadian tablets	Biblical Heb./Aram.	Hebrew MSS	Latin inscriptions	DSS transcriptions
Generation unit	subword/sign	word + char.	WordPiece	character sequence	word span + char. ablation
Natural lacuna eval.	expert sample	no	yes	yes	QD literature agreement
Unknown length	greedy masks	masked spans	missing word	arbitrary length	primary protocol
Material evidence	transcription	mask extent	transcription	image + text	traces; editor-length oracle
Primary caution	domain/task scope	synthetic damage	token alignment	edition-derived targets	no physical truth

and artifact hashes; a machine-readable table specification records evidence class and sources. The snapshots reproduce the 300-span baselines, three matched ByT5 evaluations, three matched QD MLM seeds, context sensitivity, and composition exclusion. The preserved corpus validator enforces non-biblical membership, removal of modern reconstruction text, anonymous gap representation, and disjoint scrolls. Separate audits record composition overlap, exact duplicates, five-gram similarity, and exact QD/training parallels.

The corpus manifest records primary-population counts and per-scroll preserved/damaged statistics for every extreme-fragmentation flag. Target-level predictions include candidate counts, evidence-condition ranks, and attributed readings, allowing every aggregate to be recomputed without rerunning a model.

The artifact records the ETCBC Text-Fabric version, QD snapshot date, source licenses, checkpoint hashes, training epochs, batch sizes, learning rates, beam sizes, seeds, software environment, and sample hashes. Experiments ran locally on an Apple M4 Max with 36 GB unified memory using MPS. Public redistribution remains subject to source-corpus and checkpoint licenses.

10 Limitations and Responsible Use

Synthetic damage supplies observable answers but differs from natural deterioration and editorial uncertainty. QD targets cover selected disputed sites, not a representative random sample, and its attributed proposals are incomplete. Text-Fabric editorial labels can encode modern reconstructions and therefore cannot serve as physical truth. Transcription-derived traces inherit editorial judgments. Period, dialect, genre, scribal practice, formulaic language, and orthography may all confound apparent model quality.

Eligibility thresholds also induce selection effects. Requiring 20 preserved words per chunk favors scrolls with sufficiently dense local text, while a contiguous-run rule favors uninterrupted passages and suppresses the damaged contexts encountered at inference. The extreme-fragmentation flag is an operational audit threshold rather than a claim that a manuscript is philologically worthless. Results therefore generalize to eligible transcription contexts, not automatically to the most severely damaged physical fragments.

The composition-exclusion experiment is single-seed, removes training volume together with labels, and composition metadata are incomplete. The synthetic benchmark covers only one to three words and two context words per side. QD calibration measures agreement with incomplete proposals. The base model's pretraining corpus cannot be exhaustively audited. Character models solve alignment but have not solved exact multiword generation. Direct claims about ink require image inputs and image/transcription registration, which are absent here. A visual extension should use infrared imagery, split by physical fragment or plate, preserve imaging and mask provenance, and report image-only and fused candidate rankings separately; generated pixels must not be treated as recovered ink.

The system is intended for decision support. Every candidate should retain its model score, evidence condition, rejected constraints, provenance, and uncertainty. Suggestions must not be silently inserted into editions, used to overwrite attributed readings, or presented as recovered historical text. Scholars remain responsible for paleography, material joins, syntax, genre, parallels, and publication decisions.

11 Conclusion

The method is scientifically well controlled and practically useful for one bounded role: prioritizing candidates for eligible single-word lacunae when transcription-visible traces are supplied. It is not yet a general DSS restoration system. Token alignment and granularity ensembling do not solve unknown-length multiword restoration, where exact recovery approaches zero beyond one word. Shared denominators, coverage accounting, manuscript-balanced sampling, hierarchical uncertainty, and explicit oracle labels make this mixed verdict reproducible without turning literature agreement into historical truth.

Responsible NLP Research Checklist

This draft follows the ARR checklist structure [4]. Data provenance, language, licensing, exclusions, reconstruction redaction, compute, statistical units, and limitations are stated above. This work contains no human-participant study.

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